MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, VOLTAGE COMPARATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, voltage comparators. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quad voltage comparator, single supply, low power
02	Dual voltage comparator, single supply, low power

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-14 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
G	A-1 (8-lead, can)
P	D-4 (8-lead, 1/4" x 3/8"), dual-in-line package

1.3 Absolute maximum ratings.

Positive supply voltage	36 V
Total supply voltage	36 V or ±18 V
Output voltage	36 V
Input voltage range	-0.3 V to +36 V 1/
Differential input voltage	36 V
Sink current	29 mA (approximately)
Output short circuit to ground	Continuous 2/
Storage temperature range	-65°C to +150°C
Junction temperature (T_J) 3/	+175°C
Lead temperature (soldering, 60 s)	+300°C

1/ The input voltage should be within the common mode range to insure a proper output state.
2/ No protection exists for short circuits to the positive supply.
3/ For short term test (in the specific burn-in and life test configuration when required and up to 168 hours maximum) T_J = 275°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBE-2), Griffiss AFB NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC}) - - - - - - - - - - - - 5 V dc to 30 V dc
Operating temperature range - - - - - - - - -55°C to +125°C

1.5 Power and thermal characteristics.

Case outline	Package	Maximum allowable power dissipation	Maximum θJ-C	Maximum θJ-A
A, D	14-lead FP	350 mW @ T_A = 125°C	60°C/W	140°C/W
C, P	Dual-in-line	400 mW @ T_A = 125°C	35°C/W	120°C/W
G	8-lead can	330 mW @ T_A = 125°C	40°C/W	150°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Circuit diagram and terminal connections. The circuit diagram and terminal connections shall be as specified on figure 1.

3.2.2 Functional schematic circuits. The functional schematic circuit shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

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TABLE I. Electrical performance characteristics.

Characteristics	Symbol	Conditions 5 V < $+V_{CC}$ < 30 V (see 3.4 and figure 4 unless otherwise indicated)		Limits Min	Limits Max	Unit
Input offset voltage	V_{IO}	1/	T_A = 25°C	-5	5	mV
			-55°C ≤ T_A ≤ 125°C	-7	7	mV
Input offset voltage temperature sensitivity	$\frac{\Delta V_{IO}}{\Delta T}$	-55°C ≤ ΔT_A ≤ 25°C 25°C ≤ ΔT_A ≤ 125°C		-25 -25	25 25	µV/°C "
Input offset current	I_{IO}	1/ R_S = 20 k	25°C ≤ T_A ≤ 125°C	-25	25	nA
			T_A = -55°C	-75	75	"
Input offset current temperature sensitivity	$\frac{\Delta I_{IO}}{\Delta T}$	-55°C ≤ T_A ≤ 25°C 25°C ≤ T_A ≤ 125°C		-400 -300	400 300	pA/°C "
Input bias current	$+I_{IB}$	1/ R_S = 20 k	25°C ≤ T_A ≤ 125°C T_A = -55°C	-100 -200	-1 -1	nA "
	$-I_{IB}$	1/ R_S = 20 k	25°C ≤ T_A ≤ 125°C T_A = -55°C	-100 -200	-1 -1	nA "
Input voltage common mode rejection	CMR	2/	$+V_{CC}$ = 30 V	76		dB
			$+V_{CC}$ = 5 V	70		"
Output leakage	I_{CEX}	$+V_{CC}$=V_O= +30V	-55°C ≤ T_A ≤ 125°C		1.0	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions 5 V < $+V_{CC}$ < 30 V (see 3.4 and figure 4, unless otherwise indicated)			Limits Min	Limits Max	Units
Input leakage current	$+I_{IL}$	$+V_{CC}$ = 36 V, V_{+I} = 34 V, V_{-I} = 0 V			-500	500	nA
	$-I_{IL}$	$+V_{CC}$ = 36 V, V_{+I} = 0 V, V_{-I} = 34 V			-500	500	"
Low level output voltage	V_{OL}	$+V_{CC}$ = 4.5 V	I_O = 4 mA	T_A = 25°C	---	0.4	V
			I_O = 8 mA	T_A = 25°C	---	1.5	"
			I_O = 4 mA	-55°C≤T_A≤125°C	---	0.7	"
			I_O = 8 mA	-55°C≤T_A≤125°C	---	2.0	"
Power supply current	I_{CC}	V_{ID} = 15 mV	$+V_{CC}$ = 5 V	T_A = -55°C	---	3	mA
				T_A = 25°C	---	2	"
				T_A = 125°C	---	2	"
			$+V_{CC}$ = 30 V	T_A = -55°C	---	4	"
				T_A = 25°C	---	3	"
				T_A = 125°C	---	3	"
Open loop voltage gain	A_{VS}	$+V_{CC}$ = 15 V, R_L = 15 kΩ, 1 V ≤ V_O ≤ 11 V		T_A = 25°C	50	---	V/mV
				-55°C ≤ T_A ≤ 125°C	25	---	"
Channel separation	CS	$+V_{CC}$ = 30 V, T_A = 25°C figure 8			80	---	dB
Response time low-to-high level	t_{RLH}	$+V_{CC}$ = 5 V, V_{IN} = 100 mV, R_L = 5.1 kΩ	V_{OD} = 5 mV	-55°C ≤ T_A ≤ 25°C	---	3	µs
				T_A = 125°C	---	7	"
			V_{OD} = 50 mV	-55°C ≤ T_A ≤ 25°C	---	0.8	"
				T_A = 125°C	---	1.2	"

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions $5\ V < +V_{CC} < 30\ V$ (see 3.4 and figure 4, unless otherwise indicated)			Limits		Units
					Min	Max	
Response time high-to-low level	t_{RHL}	$+V_{CC}$ = 5 V	V_{OD} = 5 mV	$-55°C \le T_A \le 25°C$	---	2.5	µs
		V_{IN} = 100 mV		$T_A = 125°C$	---	3	"
		R_L = 5.1 kΩ	V_{OD} = 50 mV	$-55°C \le T_A \le 25°C$	---	0.8	"
				$T_A = 125°C$	---	1.0	"
Voltage latch (high level input)	V_{LAT}	$+V_{CC}$ = 5 V, $T_A = 25°C$, V_{IN} = 10 V, figure 9, I_O = 4 mA			---	0.4	V

NOTES:

1/ V_{IO}, I_{IO} and $\pm I_{IB}$ shall be measured at the common mode extremes at 30 V and 5 V, as shown below:

Condition	Common mode	$+V_{CC}$	$-V_{CC}$	V_{IN}	V_O
1	(-CM)	30 V	0 V	0 V	15 V
2	(+CM)	2 V	-28 V	0 V	-13 V
3	(-CM)	5 V	0 V	0 V	1.4 V
4	(+CM)	2 V	3 V	0 V	-1.6 V

2/ CMR shall be calculated from V_{IO} measurements defined in note 1/.

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3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.5 herein.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range. Unless otherwise specified, source resistance is 50 ohms for all tests.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 50 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,4	1*,2,3,4
Group A test requirements (method 5005)	1,2,3,4, 5,6,7, 8,9	1,2,3,4, 5,6,7, 8,9
Group C end-point and group B, class S, electrical parameters (method 5005)	1,2,3 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	---
Group D end-point electrical parameters (method 5005)	1,2,3	1

*PDA applies to subgroup 1 (see 4.3d).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, except as modified herein. When the accelerated test conditions (T_A = 200°C or 250°C) are used for any test on a given inspection lot, they shall be used for all pertinent product assurance tests (screening (4.3), and group C inspection (4.4.3)) made on the lot.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

1. For class S devices: Test condition D using the circuit shown on figure 3.

2. For class B devices: Test condition D using the circuit shown on figure 3, test condition C using the circuit shown on figure 4, or test condition F using the circuit shown on figure 5.

NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). For class S devices only using the circuit shown on figure 4.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) - The PDA for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 10 and 11 shall be omitted.

b. Electrical tests requirements shall be as specified in table II herein.

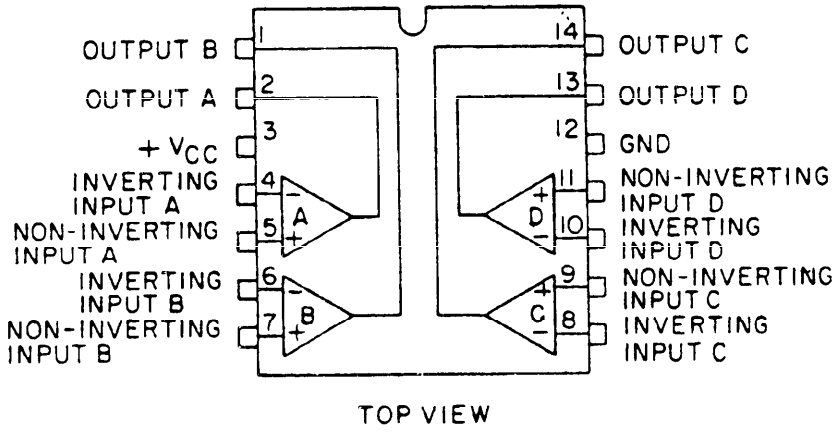
4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class S (method 1005 of MIL-STD-883). Test condition F using the circuit shown on figure 5 (see 4.3a.2 note). If the alternate burn-in conditions are used, the circuit shown on figure 4 shall be used.

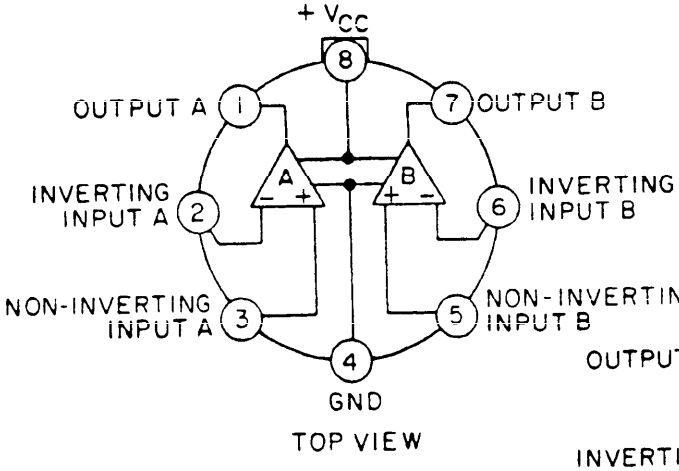
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Device type 01
Cases A, C, and D



Device type 02

Case G



Case P

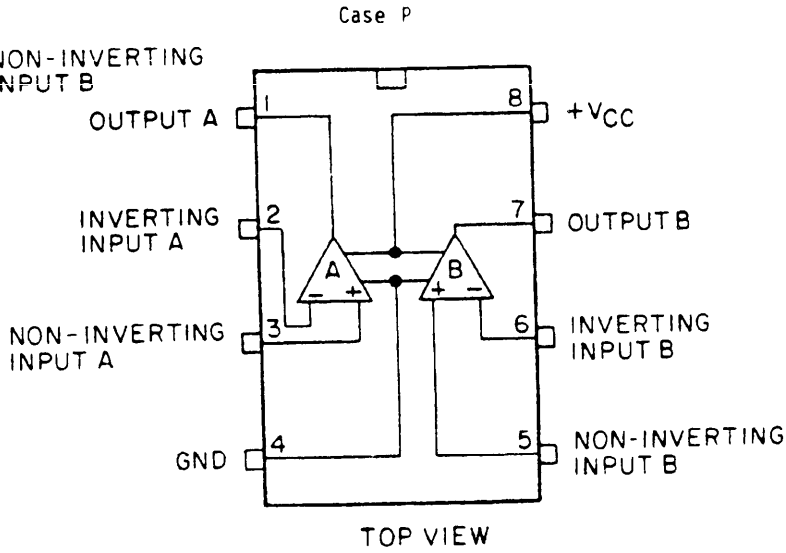
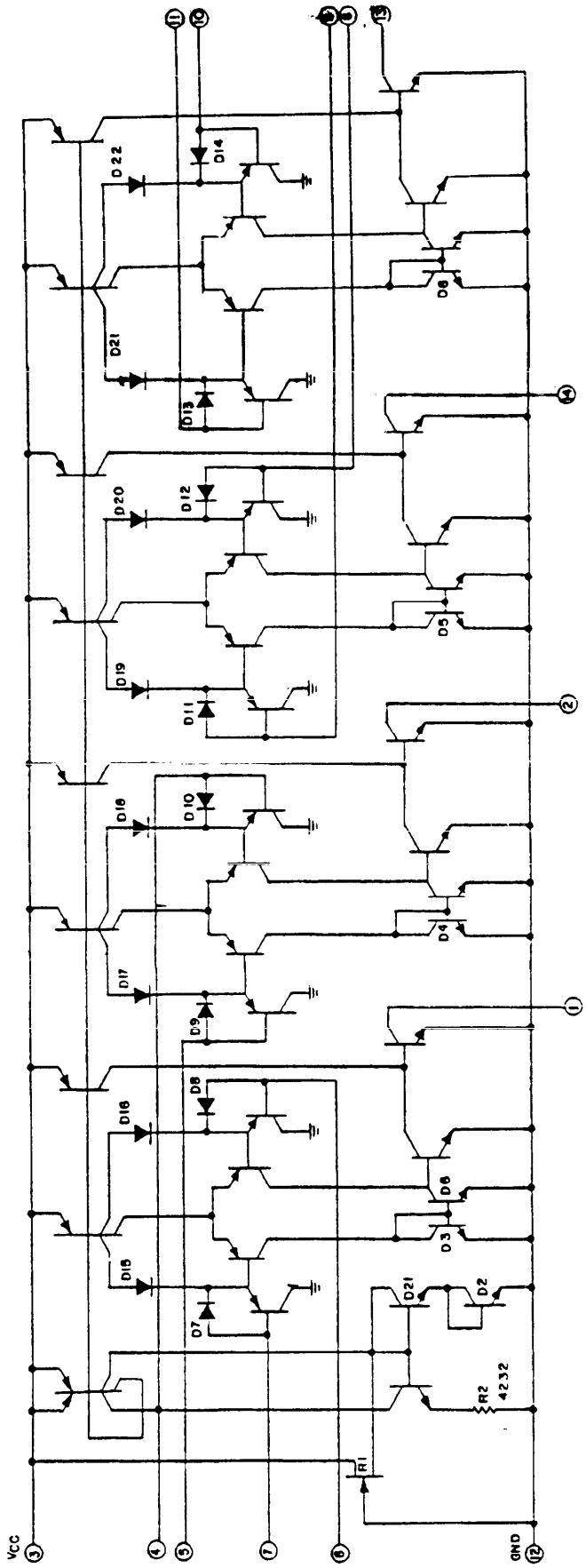


FIGURE 1. Terminal connections.

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Device types 01 and 02
(1/4 device type 01, 1/2 device type 02)



NOTE: All resistor values are in Ω.

FIGURE 2. Functional schematic circuits.

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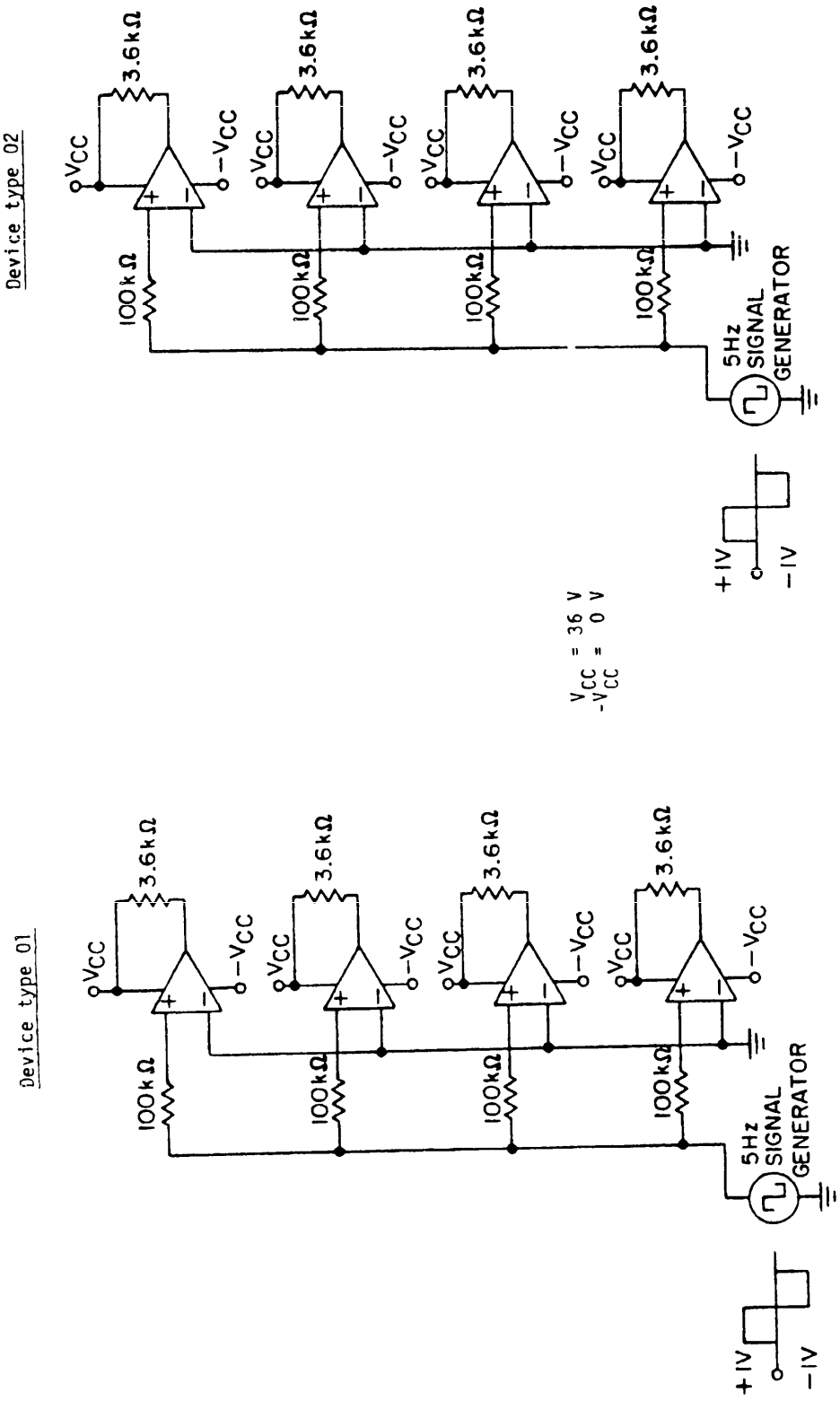


FIGURE 3. Test circuit, burn-in and steady state life test.

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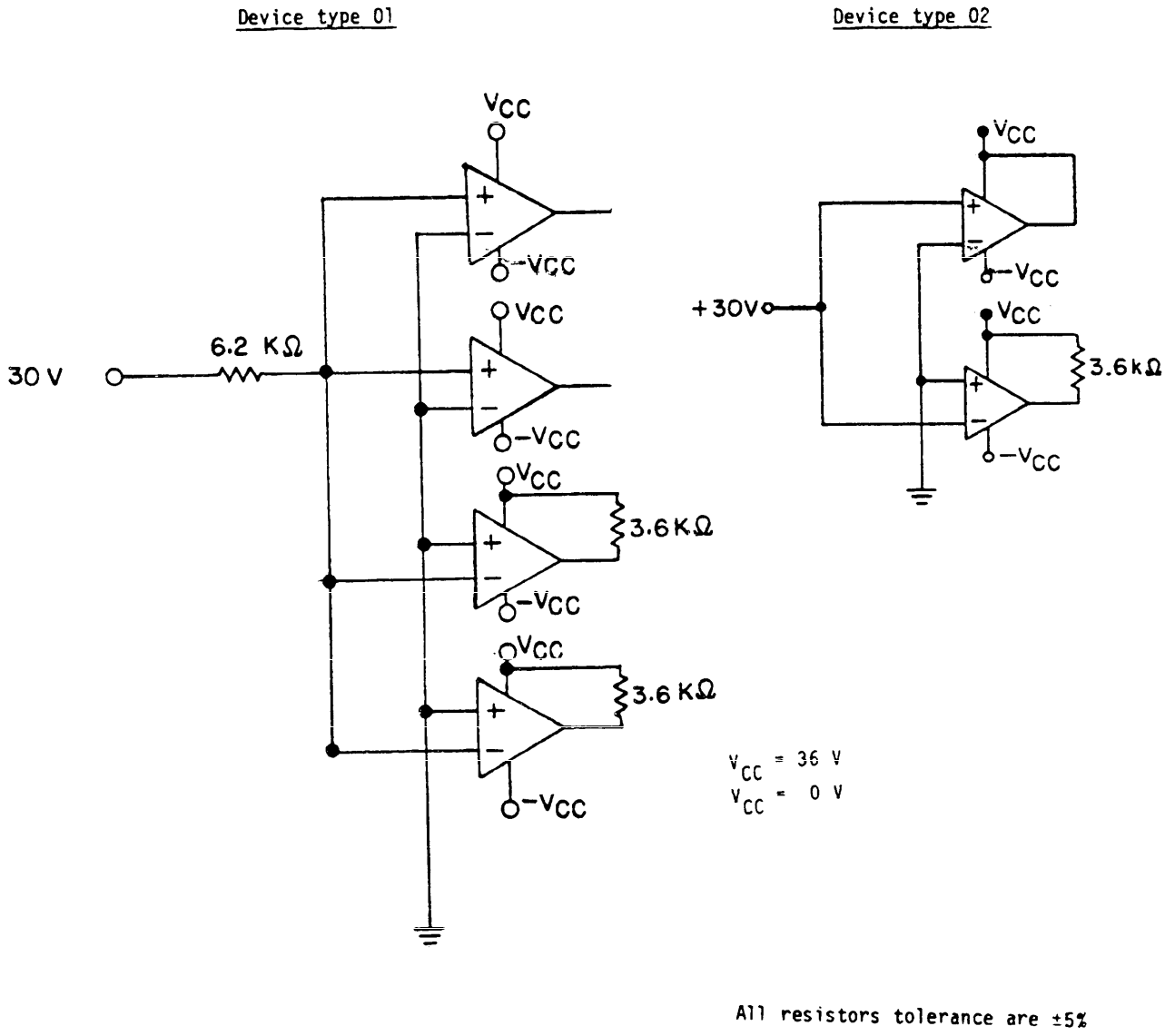
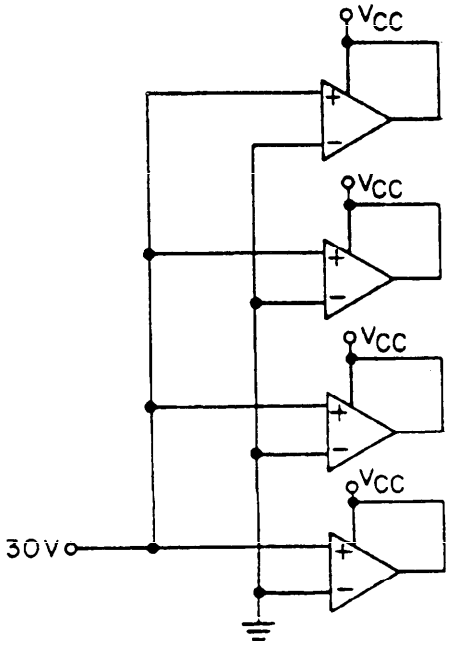


FIGURE 4. Test circuit, burn-in (steady state power and reverse bias) and steady state life test.

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Device type 01



V_{CC} = 36 V
$-V_{CC}$ = 0 V

NOTE: If accelerated, high temperature test conditions are used, the device manufacturer shall ensure that at least 85% of the applied voltage is dropped across the device under test at temperature. The device is not considered functional under accelerated test conditions.

Device type 02

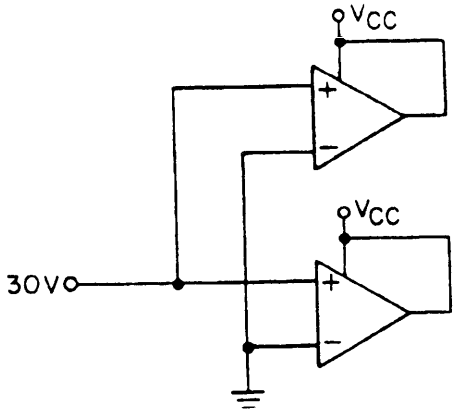
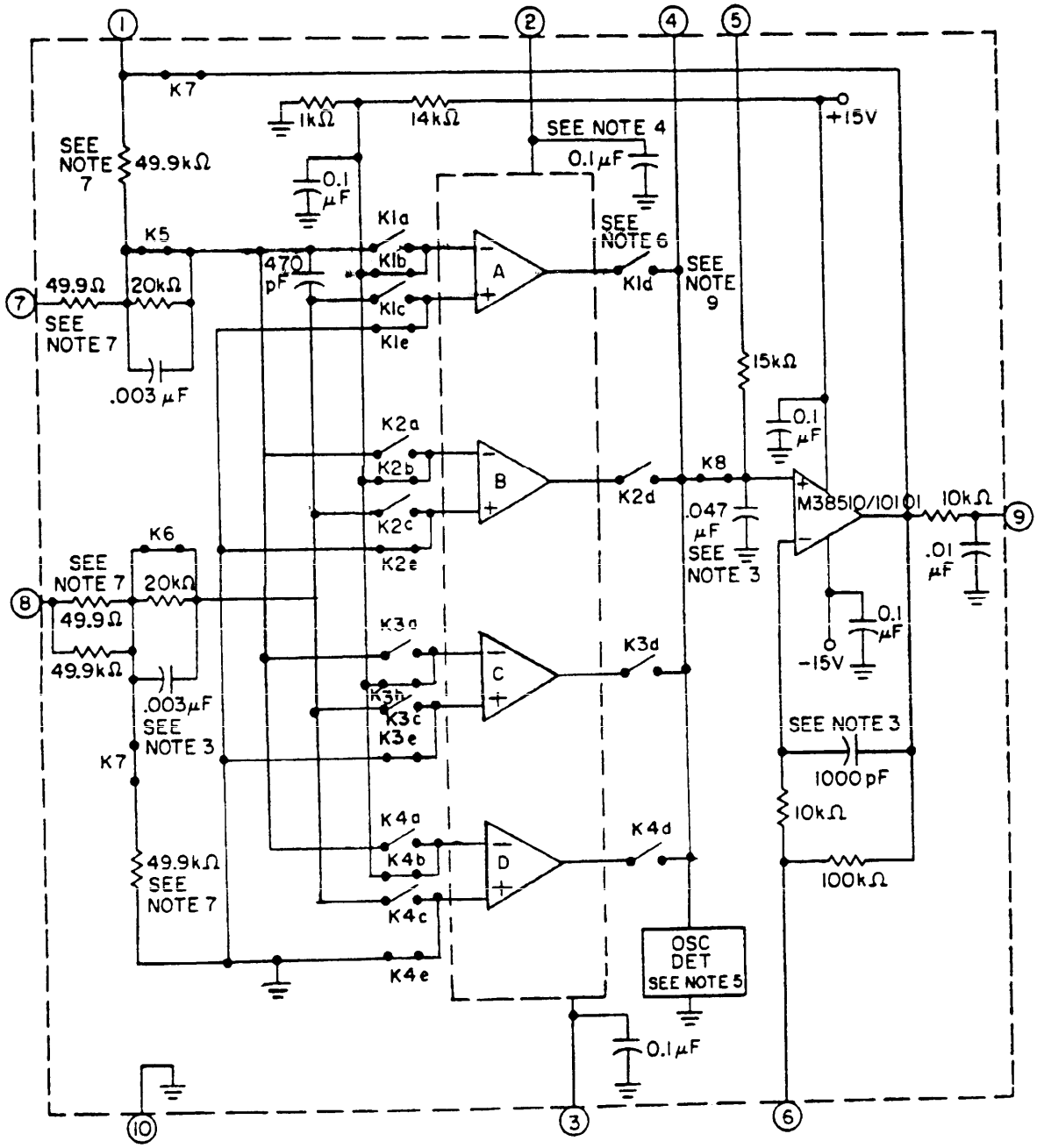


FIGURE 5. Accelerated burn-in and life test circuits.

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Device type 01

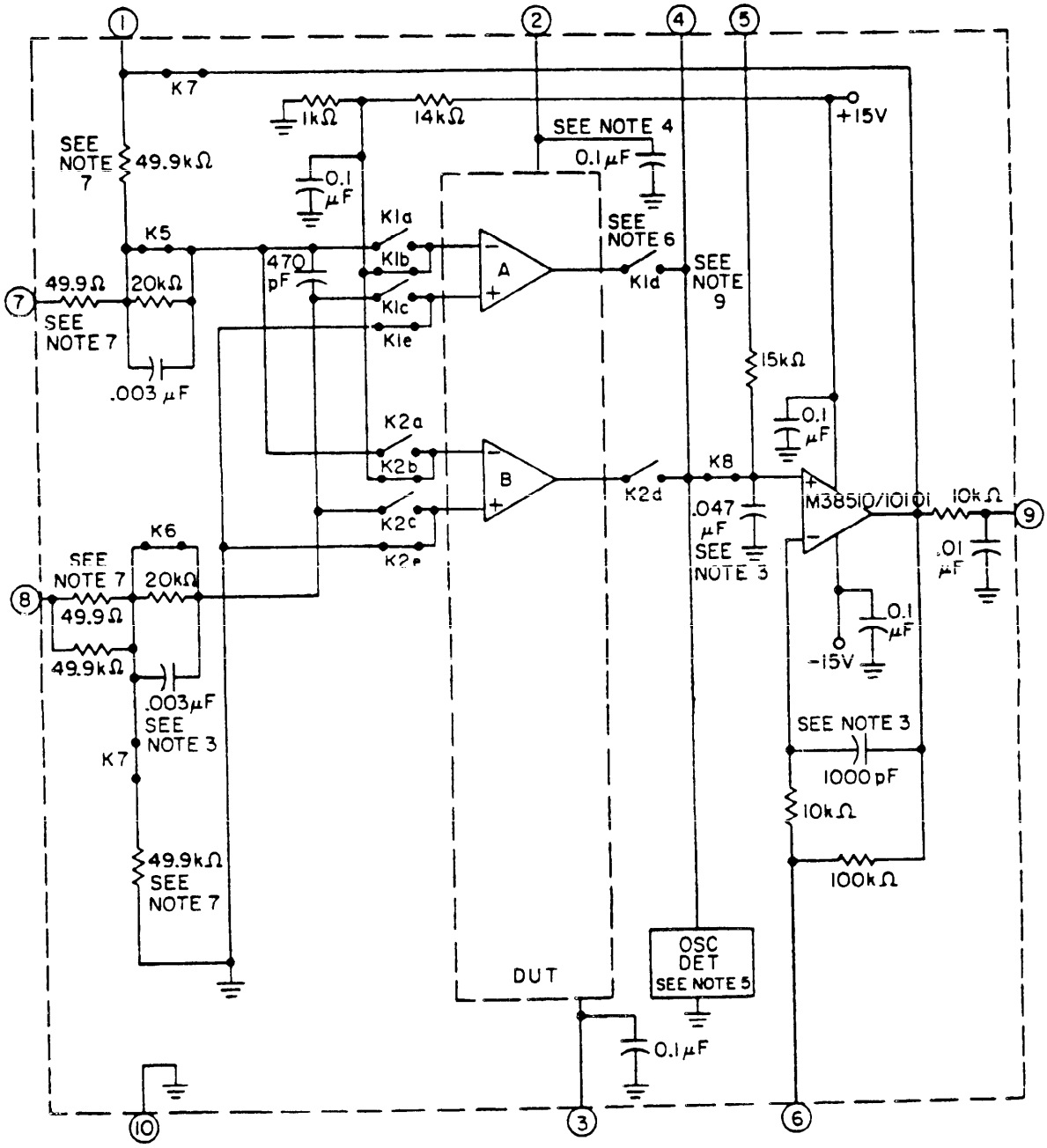


See following notes.

FIGURE 6. Test circuit for static and dynamic tests.

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Device type 02



See following notes.

FIGURE 6. Test circuit for static and dynamic tests - Continued.

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NOTES:
1. Test circuit pin conditions shall be as specified in the schedule of this figure.
2. Subgroups, test temperatures and limits shall be as specified in table III.
3. As required to prevent oscillations. Also, proper wiring procedures shall be followed to prevent oscillations. Loop response and settling time shall be consistant with test rate such that any value has settled to within 5% of its final value before measuring. Suggested values shown may not ensure loop stability for all layouts. Actual compensation also shall be approved by preparing activity prior to use.
4. Precautions shall be taken to prevent damage to the D.U.T. during insertion into the socket and change of relay contacts.
5. Any oscillation greater than 300 mV (pk-pk) shall be cause for device failure.
6. Relays K1 thru K4 select the comparator under test. The idle comparators have 1V applied to the (-) input to force their outputs to the low state.
7. These resistors are ±0.1% tolerance matched to ±.01%. All other resistors are ±1% tolerance and capacitors are 10% tolerance.
8. Common mode rejection is calculated using the offset voltage values measured at the common mode range end points.
9. The relays shown indicate test connections only. All relays are shown in their de-energized states. Relay coils are not shown.
10. Saturation of the nulling amplifier is not allowed on tests where E value is measured.

FIGURE 6. Test circuit for static and dynamic tests - Continued.

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Device type 01

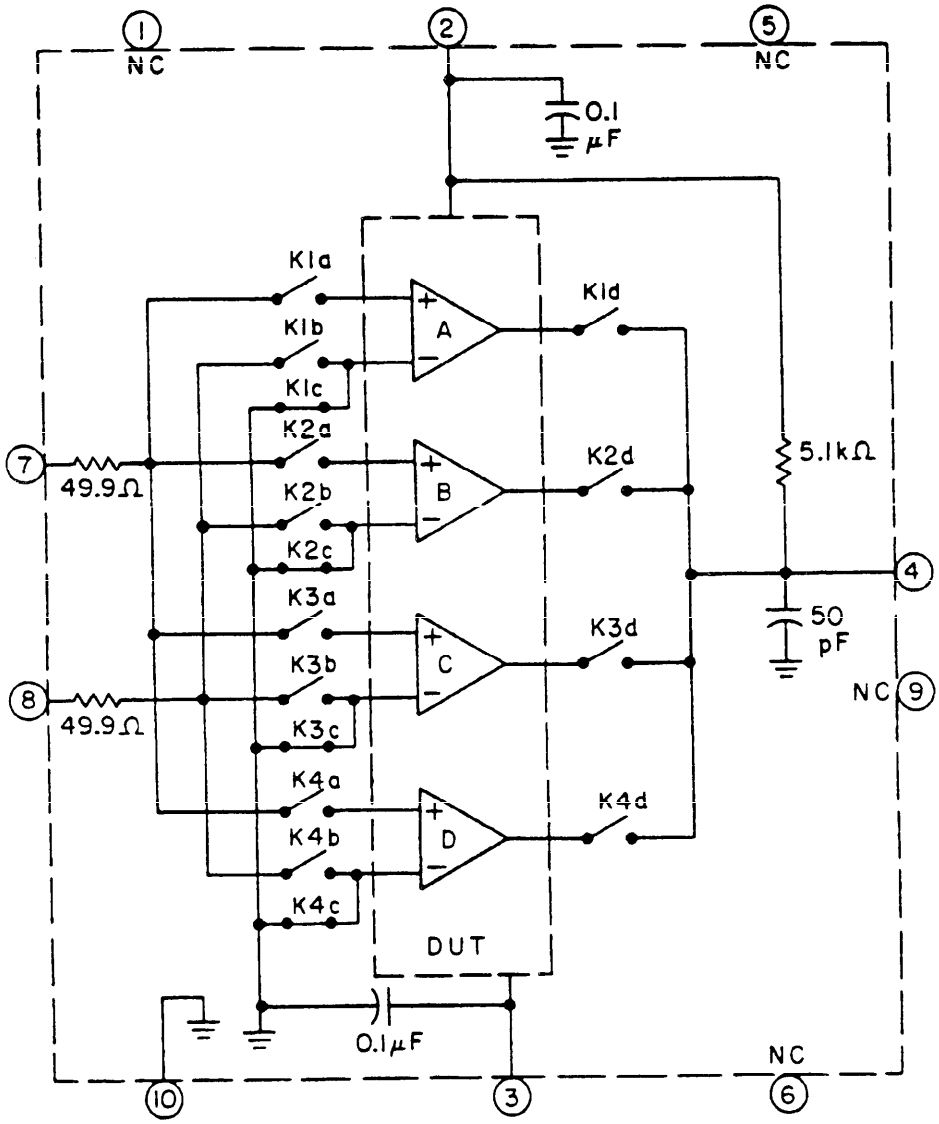


FIGURE 7. Response time test circuit and waveforms.

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Device type 02

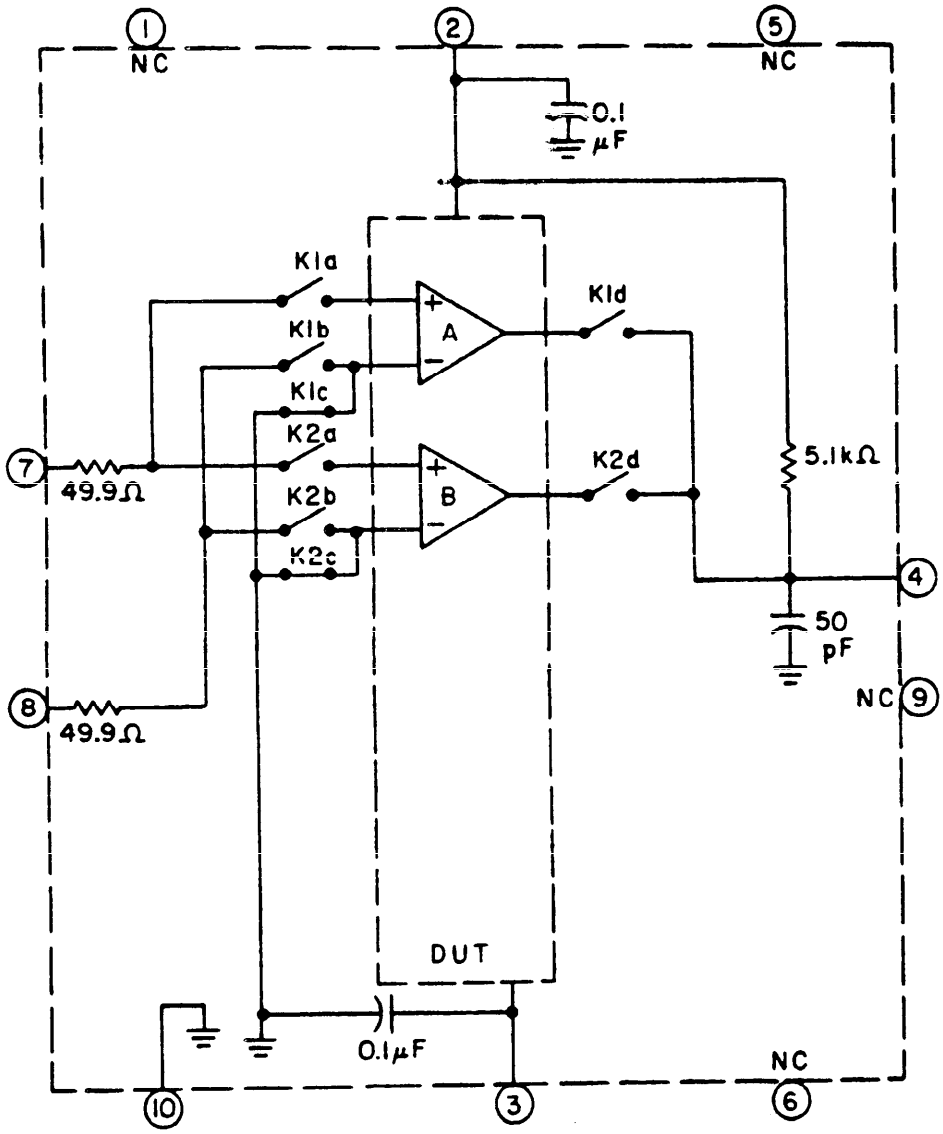
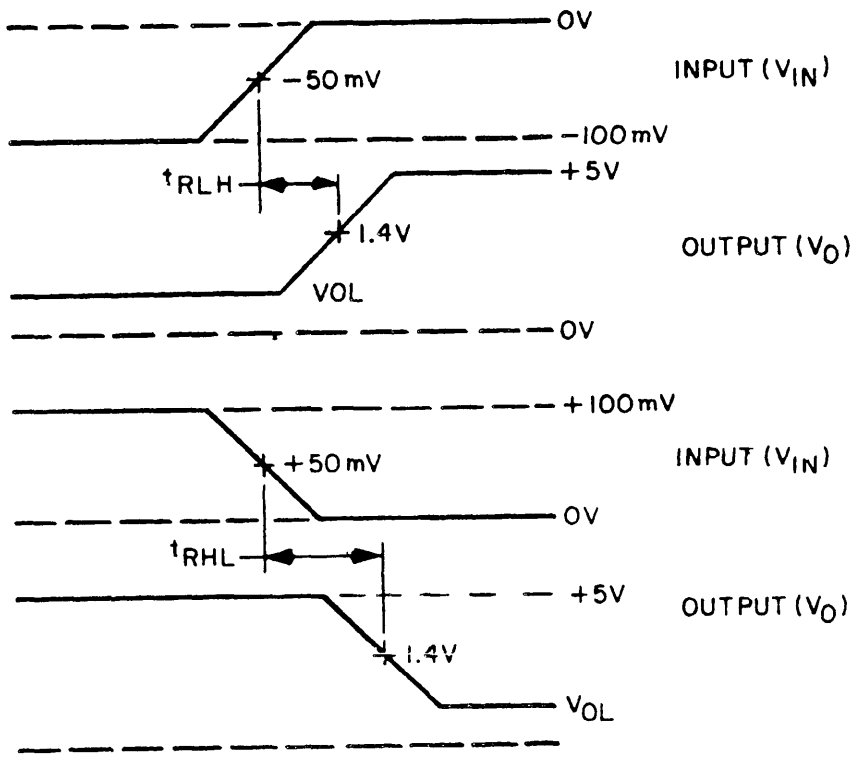


FIGURE 7. Response time test circuit and waveforms - Continued.

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NOTES:
1. Adjust the signal generator so that V_{IN} is a 100 mV pulse train with a 10 µs pulse width at 50 kHz, t_{TLH} and $t_{THL} \leq 10$ ns and $Z_O \approx 50\Omega$.
2. Setup procedure:
   a. With $V_{IN} = 0$, adjust V_{ref} from -5.0 mV to +5.0 mV in 0.1 mV steps and stop when output switches from high to low.
   b. Change V_{ref} from the value obtained in (a) above by the required V_{OD} (overdrive).
   c. Apply V_{IN} and measure the response time.
3. All resistor tolerances are ±1% and all capacitor tolerances are ±10%.
4. The output capacitance includes scope, probe and jig capacitance.

FIGURE 7. Response time test circuit and waveforms - Continued.

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Device type 01

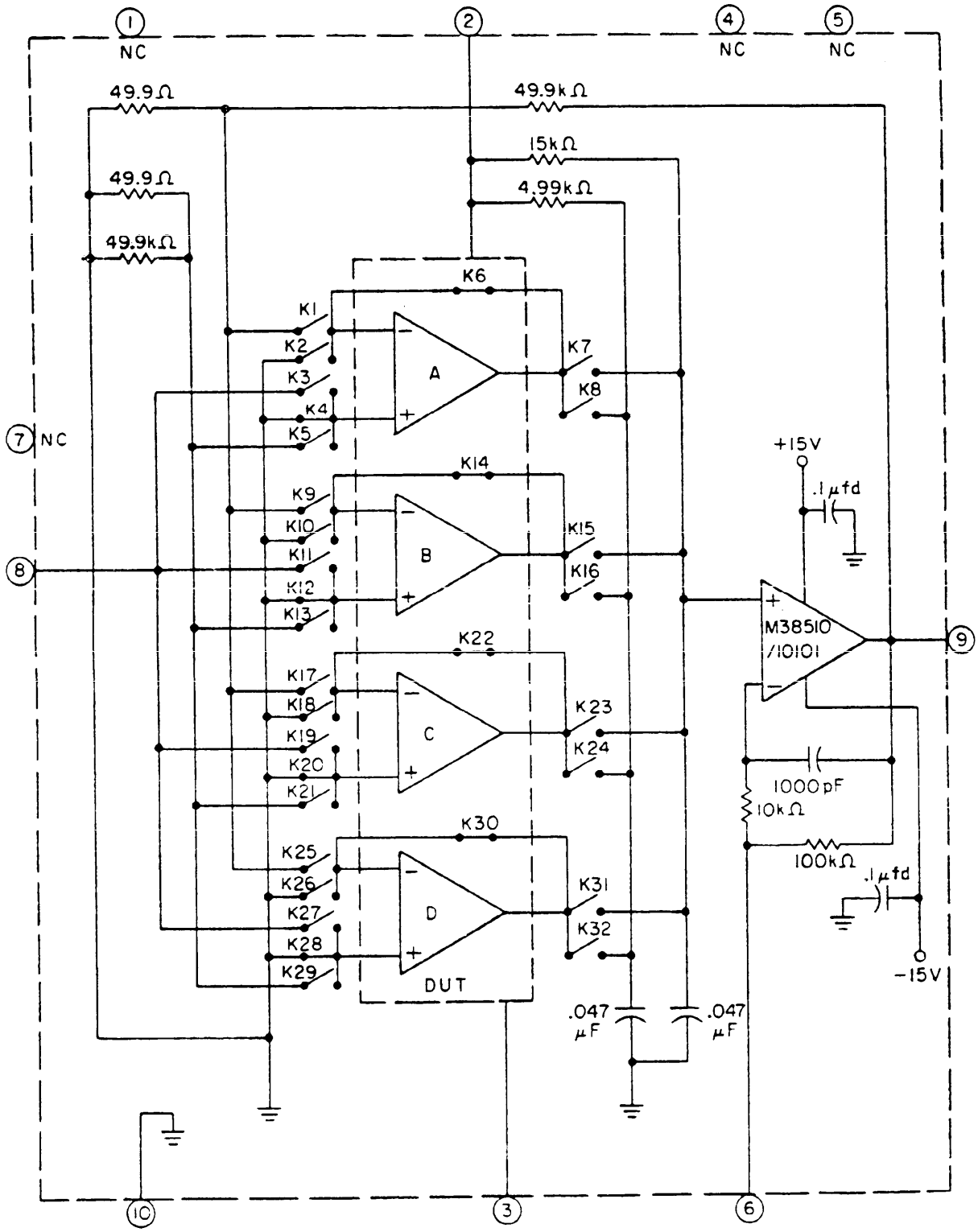


FIGURE 8. Channel separation test circuit.

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Device type 02

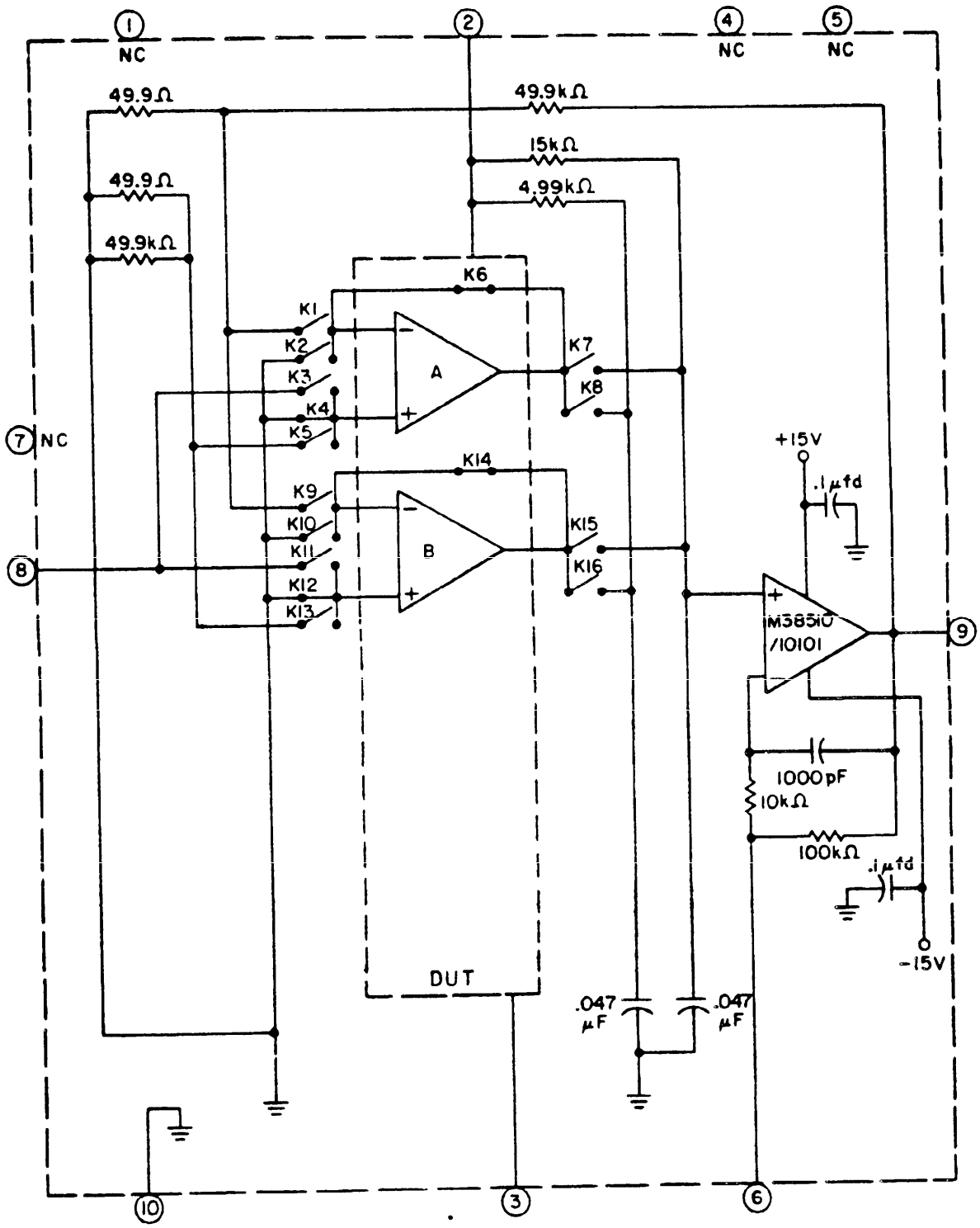


FIGURE 8. Channel separation test circuit - Continued.

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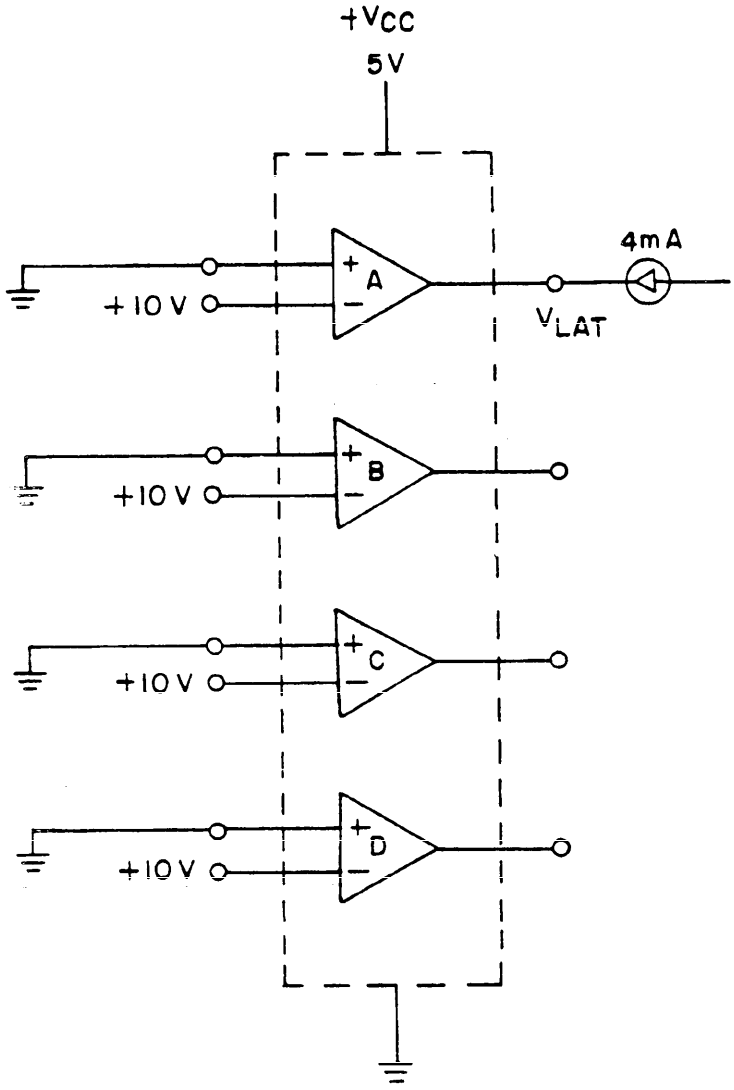
NOTES:
1. All resistors are ±0.1% tolerance and all capacitors are ±10% tolerance.
2. Precautions shall be taken to prevent damage to the D.U.T. during insertion into socket and change of state of relays (i.e. disable voltage supplies, current limit $\pm V_{CC}$, etc.).
3. All relays are shown in the normal de-energized state. The following table shall be used to determine which relays to energize for each test. (For device type 02, use tests 87 and 90 only.)

Channel sep. test no.	Channels tested	Relays energized	
		Driven channel	Monitored channel
87	A to B	2,3,4,6,8	9,12,13,14,15
88	A to C		17,20,21,22,23
89	A to D		25,28,29,30,31
90	B to A	10,11,12,14,16	1,4,5,6,7
91	B to C		17,20,21,22,23
92	B to D		25,28,29,30,31
93	C to A	18,19,20,22,24	1,4,5,6,7
94	C to B		9,12,13,14,15
95	C to D		25,28,29,30,31
96	D to A	26,27,28,30,32	1,4,5,6,7
97	D to B		9,12,13,14,15
98	D to C		17,20,21,22,23

4. The nulling amplifier shall be a JM38510/10101XXX. Saturation of the nulling amplifier is not allowed.

FIGURE 8. Channel separation test circuit - Continued.

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NOTES:
1. Test shall be performed on each comparator separately (device type 01 - four comparators, device type 02 - two comparators).
2. Constant current source shall supply 4 mA into output.

FIGURE 9. Voltage latch test circuit.

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TABLE III. Group A inspection.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Measured pin Value	Measured pin Unit	Equation	Limits Min	Limits Max	Unit
1 T_A = 25°C	V_{IO}	4001	1	1/ 11/		30 V	0 V		30 V	15 V	0 V	0 V		GND	10/	9	E_1	V	$V_{IO} = E_1$	-5	5	mV
	"	"	2	"		2 V	-28 V		2 V	-13 V	"	"		"	10/	"	E_2	"	$V_{IO} = E_2$	"	"	"
	"	"	3	"		5 V	0 V		5 V	1.4 V	"	"		"	10/	"	E_3	"	$V_{IO} = E_3$	"	"	"
	"	"	4	"		2 V	-3 V		2 V	-1.6 V	"	"		"	10/	"	E_4	"	$V_{IO} = E_4$	"	"	"
	I_{IO}	"	5	"		30 V	0 V		30 V	15 V	"	"		"	K5, K6 10/	"	E_5	"	$I_{IO} = 50 (E_1 - E_5)$	-25	25	nA
	"	"	6	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_6	"	$I_{IO} = 50 (E_2 - E_6)$	"	"	"
	"	"	7	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_7	"	$I_{IO} = 50 (E_3 - E_7)$	"	"	"
	"	"	8	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_8	"	$I_{IO} = 50 (E_4 - E_8)$	"	"	"
	$+I_{IB}$	"	9	"		30 V	0 V		30 V	15 V	"	"		"	K6 10/	"	E_9	"	$+I_{IB} = 50 (E_1 - E_9)$	-100	-1	"
	"	"	10	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{10}	"	$+I_{IB} = 50 (E_2 - E_{10})$	"	"	"
	"	"	11	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{11}	"	$+I_{IB} = 50 (E_3 - E_{11})$	"	"	"
	"	"	12	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{12}	"	$+I_{IB} = 50 (E_4 - E_{12})$	"	"	"
	$-I_{IB}$	"	13	"		30 V	0 V		30 V	15 V	"	"		"	K5 10/	"	E_{13}	"	$-I_{IB} = 50 (E_{13} - E_{11})$	"	"	"
	"	"	14	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{14}	"	$-I_{IB} = 50 (E_{14} - E_2)$	"	"	"
	"	"	15	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{15}	"	$-I_{IB} = 50 (E_{15} - E_3)$	"	"	"
	"	"	16	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{16}	"	$-I_{IB} = 50 (E_{16} - E_4)$	"	"	"
	CMR	4003	17	2/	Calculate value using data from tests 1 and 2 8/														$CMR = 20 \log \left\lvert \frac{28000}{E_1 - E_2} \right\rvert$	75		dB
	CMR	4003	18	2/	Calculate value using data from tests 3 and 4 8/														$CMR = 20 \log \left\lvert \frac{30000}{E_3 - E_4} \right\rvert$	70		dB
	I_{CEX}	3009	19	3/	-15 V	30 V	0 V	30 V	0 V	0 V	0 V	0 V		GND	K7, K8 10/	4	I_1	mA	$I_{CEX} = I_1$		1.0	µA
	$+I_{IL}$	4001	20	7/		36 V	"		"	"	0 V	34 V		"	K7, K8 10/	8	I_2	nA	$+I_{IL} = I_2$	-500	500	nA
	$-I_{IL}$	4001	21	7/		36 V	"		"	"	34 V	0 V		"	K7, K8 10/	7	I_3	nA	$+I_{IL} = I_3$	-500	500	nA
	V_{OL}	3007	22	I_O = 4 mA	15 V	4.5 V	"	4 mA	"	"	0 V	0 V		"	K7, K8 10/	6	E_{17}	V	$V_{OL} = E_{17}$		0.4	V
	V_{OL}	3007	23	I_O = 8 mA	15 V	4.5 V	"	8 mA	"	"	"	"		"	K7, K8 10/	6	E_{18}	V	$V_{OL} = E_{18}$		1.5	V
	I_{CC}	3005	24	$V_O = V_{OL}$ 4/	15 V	5 V	"		"	"	"	"		"	K1,K2,K3,K4,K7,K8	2	I_4	mA	$I_{CC} = I_4$		2	mA
	I_{CC}	3005	25	$V_O = V_{OL}$ 4/	15 V	30 V	"		"	"	"	"		"	K1,K2,K3,K4,K7,K8	2	I_5	mA	$I_{CC} = I_5$		3	mA
2 T_A = 125°C	V_{IO}	4001	26	1/		30 V	0 V		30 V	15 V	"	"		"	10/	9	E_{19}	V	$V_{IO} = E_{19}$	-7	7	mV
	"	"	27	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{20}	"	$V_{IO} = E_{20}$	"	"	"
	"	"	28	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{21}	"	$V_{IO} = E_{21}$	"	"	"
	"	"	29	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{22}	"	$V_{IO} = E_{22}$	"	"	"
	$\frac{\Delta V_{IO}}{\Delta t}$		30	6/	$\frac{\Delta V_{IO}}{\Delta t} = \left[V_{IO} \text{ (test 26)} - V_{IO} \text{ (test 1)} \right] / 100°C$															-25	25	µV/°C
	I_{IO}	4001	31	1/		30 V	0 V		30 V	15 V	0 V	0 V		GND	K5, K6 10/	9	E_{23}	V	$I_{IO} = 50 (E_{19} - E_{23})$	"	"	nA
	"	"	32	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{24}	"	$I_{IO} = 50 (E_{20} - E_{24})$	"	"	"
	"	"	33	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{25}	"	$I_{IO} = 50 (E_{21} - E_{25})$	"	"	"
	"	"	34	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{26}	"	$I_{IO} = 50 (E_{23} - E_{26})$	"	"	"
	$\frac{\Delta I_{IO}}{\Delta t}$		35	6/	$\frac{\Delta I_{IO}}{\Delta t} = \left[I_{IO} \text{ (test 31)} - I_{IO} \text{ (test 5)} \right] / 100°C$															-300	300	pA/°C
	$+I_{IB}$	"	36	1/		30 V	0 V		30 V	15 V	0 V	0 V		GND	K6 10/	9	E_{27}	V	$+I_{IB} = 50 (E_{19} - E_{27})$	-100	-1	nA
	"	"	37	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{28}	"	$+I_{IB} = 50 (E_{20} - E_{28})$	"	"	"
	"	"	38	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{29}	"	$+I_{IB} = 50 (E_{23} - E_{29})$	"	"	"
	"	"	39	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{30}	"	$+I_{IB} = 50 (E_{24} - E_{30})$	"	"	"
	$-I_{IB}$	"	40	"		30 V	0 V		30 V	15 V	"	"		"	K5 10/	"	E_{31}	"	$-I_{IB} = 50 (E_{31} - E_{19})$	"	"	"
	"	"	41	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{32}	"	$-I_{IB} = 50 (E_{32} - E_{20})$	"	"	"
	"	"	42	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{33}	"	$-I_{IB} = 50 (E_{33} - E_{21})$	"	"	"
	"	"	43	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{34}	"	$-I_{IB} = 50 (E_{34} - E_{22})$	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Measured pin Value	Measured pin Unit	Equation	Limits Min	Limits Max	Unit
																			$CMR = 20 \log \frac{28000}{\lvert E_{19}-E_{20} \rvert}$	76		dB
																			$CMR = 20 \log \frac{30000}{\lvert E_{21}-E_{22} \rvert}$	70		dB
2	CMR	4003	44	2/	Calculate value using data from tests 26 and 27 8/																	
T_A = 125°C	CMR	4003	45	2/	Calculate value using data from tests 28 and 29 8/																	
	I_{CEX}	3009	46	3/	-15 V	30 V	0 V	30 V	0 V	0 V	0 V	0 V		GND	K7, K8 10/	4	I_6	µA	$I_{CEX} = I_6$		1.0	µA
						36 V	"		"	"	0 V	34 V		"	K7, K8 10/	8	I_7	nA	$+I_{IL} = I_7$	-500	500	nA
	$+I_{IL}$	4001	47	7/		36 V	"		"	"	34 V	0 V		"	K7, K8 10/	7	I_8	nA	$-I_{IL} = I_8$	-500	500	nA
	$-I_{IL}$	4001	48	7/																		
	V_{OL}	3007	49	I_O = 4 mA	15 V	4.5 V	"	4 mA	"	"	0 V	0 V		"	K7, K8 10/	4	E_{35}	V	$V_{OL} = E_{35}$		0.7	V
	V_{OL}	3007	50	I_O = 8 mA	"	4.5 V	"	8 mA	"	"	"	"		"	K7, K8 10/	4	E_{36}	V	$V_{OL} = E_{36}$		2.0	V
	I_{CC}	3005	51	$V_O = V_{OL}$ 4/	"	5 V	"		"					"	K1,K2,K3,K4,K7,K8	2	I_9	mA	$I_{CC} = I_9$		2	mA
	I_{CC}	3005	52	$V_O = V_{OL}$ 4/	"	30 V	"		"					"	K1,K2,K3,K4,K7,K8	2	I_{10}	mA	$I_{CC} = I_{10}$		3	mA
3	V_{IO}	4031	53	1/		30 V	0 V		30 V	15 V	"	"		"	10/	9	E_{37}	V	$V_{IO} = E_{37}$	-7	7	mV
T_A = -55°C	"	"	54	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{38}	"	$V_{IO} = E_{38}$	"	"	"
	"	"	55	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{39}	"	$V_{IO} = E_{39}$	"	"	"
	"	"	56	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{40}	"	$V_{IO} = E_{40}$	"	"	"
	$\frac{\Delta V_{IO}}{\Delta t}$	-	57	6/	$\frac{\Delta V_{IO}}{\Delta t}$ = [V_{IO} (test 53) − V_{IO} (test 1)] /80°C															-25	25	µV/°C
	I_{IO}	"	58	1/		30 V	0 V		30 V	15 V	0 V	0 V		GND	K5, K6 10/	9	E_{41}	V	$I_{IO} = 50 (E_{37} - E_{41})$	-75	75	nA
	"	"	59	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{42}	"	$I_{IO} = 50 (E_{38} - E_{42})$	"	"	"
	"	"	60	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{43}	"	$I_{IO} = 50 (E_{39} - E_{43})$	"	"	"
	"	"	61	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{44}	"	$I_{IO} = 50 (E_{40} - E_{44})$	"	"	"
	$\frac{\Delta I_{IO}}{\Delta t}$	-	62	6/	$\frac{\Delta I_{IO}}{\Delta t}$ = [I_{IO} (test 58) − I_{IO} (test 5)] /80°C															-400	400	pA/°C
	$+I_{IB}$	"	63	1/		30 V	0 V		30 V	15 V	0 V	0 V		GND	K6 10/	9	E_{45}	V	$+I_{IB} = 50 (E_{37} - E_{45})$	-200	-1	nA
	"	"	64	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{46}	"	$+I_{IB} = 50 (E_{38} - E_{46})$	"	"	"
	"	"	65	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{47}	"	$+I_{IB} = 50 (E_{39} - E_{47})$	"	"	"
	"	"	66	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{48}	"	$+I_{IB} = 50 (E_{40} - E_{48})$	"	"	"
	$-I_{IB}$	"	67	"		30 V	0 V		30 V	15 V	"	"		"	K5 10/	"	E_{49}	"	$-I_{IB} = 50 (E_{49} - E_{37})$	"	"	"
	"	"	68	"		2 V	-28 V		2 V	-13 V	"	"		"	"	"	E_{50}	"	$-I_{IB} = 50 (E_{50} - E_{38})$	"	"	"
	"	"	69	"		5 V	0 V		5 V	1.4 V	"	"		"	"	"	E_{51}	"	$-I_{IB} = 50 (E_{51} - E_{39})$	"	"	"
	"	"	70	"		2 V	-3 V		2 V	-1.6 V	"	"		"	"	"	E_{52}	"	$-I_{IB} = 50 (E_{52} - E_{40})$	"	"	"
																			$CMR = 20 \log \frac{28000}{\lvert E_{37}-E_{38} \rvert}$	76		dB
																			$CMR = 20 \log \frac{30000}{\lvert E_{39}-E_{40} \rvert}$	70		dB
	CMR	4003	71	2/	Calculate value using data from tests 53 and 54 8/																	
	CMR	4003	72	2/	Calculate value using data from tests 55 and 56 8/																	
	I_{CEX}	3009	73	3/	-15 V	30 V	0 V	30 V	0 V	0 V	0 V	0 V		GND	K7, K8 10/	4	I_{11}	µA	$I_{CEX} = I_{11}$		1.0	µA
						36 V	"		"	"	0 V	34 V		"	K7, K8 10/	8	I_{12}	nA	$+I_{IL} = I_{12}$	-500	500	nA
	$+I_{IL}$	4001	74	7/		36 V	"		"	"	34 V	0 V		"	K7, K8 10/	7	I_{13}	nA	$-I_{IL} = I_{13}$	-500	500	nA
	$-I_{IL}$	4001	75	7/																		
	V_{OL}	3007	76	I_O = 4 mA	15 V	4.5 V	"	4 mA	"	"	0 V	"		"	K7, K8 10/	4	E_{53}	V	$V_{OL} = E_{53}$		0.7	V
	V_{OL}	3007	77	I_O = 8 mA	"	4.5 V	"	8 mA	"	"	"	"		"	K7, K8 10/	4	E_{54}	V	$V_{OL} = E_{54}$		2.0	V
	I_{CC}	3005	78	$V_O = V_{OL}$ 4/	"	15 V	"		"					"	K1,K2,K3,K4,K7,K8	2	I_{14}	mA	$I_{CC} = I_{14}$		3	mA
	I_{CC}	3005	79	$V_O = V_{OL}$ 4/	"	30 V	"		"					"	K1,K2,K3,K4,K7,K8	2	I_{15}	mA	$I_{CC} = I_{15}$		4	mA

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Measured pin Value	Measured pin Unit	Equation	Limits Min	Limits Max	Unit
4 T_A = 25°C	A_{VS}	4004	80	5/		15 V	0 V		15 V	11 V 1 V	0 V	0 V		GND	10/	9	E_{55} E_{56}	V	$A_V = \frac{10}{E_{56} - E_{55}}$	50		V/mV
5 T_A = 125°C	A_{VS}	4004	81	"		"	"		"	11 V 1 V	"	"		"	10/	"	E_{57} E_{58}	"	$A_V = \frac{10}{E_{58} - E_{57}}$	25		V/mV
6 T_A = -55°C	A_{VS}	4004	82	"		"	"		"	11 V 1 V	"	"		"	10/	"	E_{59} E_{60}	"	$A_V = \frac{10}{E_{60} - E_{59}}$	25		V/mV
7 T_A = 25°C	t_{RLH} t_{RLH} t_{RHL} t_{RHL}		83 84 85 86		1/ V_{CC} = 5 V; V_{IN} = 100 mV R_L = 5.1 kΩ Figure 7												V_{OD} = 5 mV V_{OD} = 50 mV V_{OD} = 5 mV V_{OD} = 50 mV	See response time waveforms		5 0.8 2.5 0.8	µs " " "
	CS		87	1/ V_{CC} = 30 V fig. 8		20 V	-10 V			15 V		1 V 1 V		GND	See figure 8	9	E_{61} E_{62}	V	$CS = 20 \log \left\lvert \frac{10000}{E_{61} - E_{62}} \right\rvert$	80		dB
	"		98	"		"	"			"		1 V 1 V		"	"	"	E_{63} E_{64}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{63} - E_{64}} \right\rvert$	"		"
	"		89	"		"	"			"		1 V 1 V		"	"	"	E_{65} E_{66}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{65} - E_{66}} \right\rvert$	"		"
	"		90	"		"	"			"		1 V 1 V		"	"	"	E_{67} E_{68}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{67} - E_{68}} \right\rvert$	"		"
	"		91	"		"	"			"		1 V -1 V		"	"	"	E_{69} E_{70}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{69} - E_{70}} \right\rvert$	"		"
	"		92	"		"	"			"		1 V -1 V		"	"	"	E_{71} E_{72}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{71} - E_{72}} \right\rvert$	"		"
	"		93	"		"	"			"		1 V -1 V		"	"	"	E_{73} E_{74}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{73} - E_{74}} \right\rvert$	"		"
	"		94	"		"	"			"		1 V -1 V		"	"	"	E_{75} E_{76}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{75} - E_{76}} \right\rvert$	"		"
	"		95	"		"	"			"		1 V 1 V		"	"	"	E_{77} E_{78}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{77} - E_{78}} \right\rvert$	"		"
	"		96	"		"	"			"		1 V -1 V		"	"	"	E_{79} E_{80}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{79} - E_{80}} \right\rvert$	"		"
	"		97	"		"	"			"		1 V -1 V		"	"	"	E_{81} E_{82}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{81} - E_{82}} \right\rvert$	"		"
	"		98	"		"	"			"		1 V -1 V		"	"	"	E_{83} E_{84}	"	$CS = 20 \log \left\lvert \frac{10000}{E_{83} - E_{84}} \right\rvert$	"		"

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	8	9	10	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
8 T_A = 125°C	t_{RLH} t_{RLH} t_{RHL} t_{RHL}		99 100 101 103		+V_{CC} = 5 V; V_{IN} = 100 mV R_L = 5.1 kΩ Figure 7												V_{OD} = 5 mV V_{OD} = 50 mV V_{OD} = 5 mV V_{OD} = 50 mV		See response time waveforms		7 1.2 3 1	µs " "
8 T_A = -55°C	t_{RLH} t_{RLH} t_{RHL} t_{RHL}		103 104 105 106		+V_{CC} = 5 V; V_{IN} = 100 mV R_L = 5.1 kΩ Figure 7												V_{OD} = 5 mV V_{OD} = 50 mV V_{OD} = 5 mV V_{OD} = 50 mV		See response time waveforms		5 0.8 2.5 0.8	" " " "
9 T_A = 25°C	V_{LAT} " "		107 108 109 110		+V_{CC} = 5 V, Figure 9 I_O = 4 mA																0.4 " " "	V " " "

1/ V_{IO}, I_{IO}, $+I_{IB}$ and $-I_{IB}$ are measured over the common mode range for 30 V and 5 V power supply conditions.

2/ CMR is calculated for 30 V and 5 V power supply conditions.

3/ I_{CCX} is measured with the output in the high state (V_{OH}) and connected to +30 V.

4/ I_{CC} is measured at no load, but with the output of each op amp in its low state.

5/ A_{VS} is measured with a 15 kΩ pullup resistor between the output levels of 1 V and 11 V.

6/ Tests 30, 35, 57 and 60 which require a read and record measurement plus a calculation may be omitted except when subgroup 2 and 3 are being accomplished for group A sampling inspection and groups C and D end point measurements.

7/ $+I_{IL}$ is measured with +34 V on the + input and 0 V on the - input. $-I_{IL}$ is measured with +34 V on the - input and 0 V on the + input.

8/ Common mode rejection is calculated using the offset voltage values measured at the common mode range end points.

9/ Each device shall be tested over the common mode range as specified in table III with the output forced to voltage midway between +V_{CC} and -V_{CC} supplies. V_{CM} is achieved by grounding the inputs and algebraically subtracting V_{CM} from each supply. Common mode rejection is calculated using offset voltage values measured at the common mode range end points.

10/ Relays K1 thru K4 select comparator under test.

11/ Each comparator shall be tested separately, except for the I_{CC} measurements where all comparators shall be connected as grounded followers.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test for class B (method 1005 of MIL-STD-883). Test condition D using the circuit shown on figure 3 or test condition F using the circuit shown on figure 5 (see 4.3a.2., note).

c. Subgroups 3 and 4 shall be added to group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 8 and 9 of table III herein.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and end-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltage value given, except the input offset voltage (or differential voltage) are referenced to the external zero reference level of the supply voltage. Currents are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and steady-state life test cooldown procedure. When devices are measured at 25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition prior to removal of the bias.

TABLE IV. Group C end-point electrical parameters (T_A = 25°C)

Test no.	Test	+VCC = 30 V			Units
		Limits		Delta	
		Min	Max		
1	V_{10}	-5	5	±1	mV
9	$+I_{IB}$	-100	-1	±15	nA
13	$-I_{IB}$	-100	-1	±15	nA

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

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c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

6.3.1 Logic threshold voltage. The approximate voltage at the output of the comparator at which the loading logic circuitry changes its digital state.

6.3.2 Voltage gain. The ratio of the change in output voltage to the change in voltage between the input terminals producing it with the dc output level in the vicinity of the logic threshold voltage.

6.3.3 Response time. The interval between the application of an input step function and the time when the output crosses the logic threshold voltage. The input step drives the comparator from some initial, saturated input voltage to an input level just barely in excess of that required to bring the output from saturation to the logic threshold voltage. This excess is referred to as the voltage overdrive.

6.3.4 Positive output level. The dc output voltage in the positive direction with the input voltage equal to or greater than a minimum specified amount.

6.3.5 Negative output level. The dc output voltage in the negative direction with the input voltage equal to or greater than a minimum specified amount.

6.3.6 Low level output voltage. The maximum low state output voltage at a specified level of sink current into the device.

6.3.7 Differential input voltage. The difference between the two voltages applied to the input terminals of an amplifier. The difference is considered positive when the non-inverting input is positive with respect to the inverting input and negative when the inverting input is positive with respect to the non-inverting input (V_{ID}).

6.3.8 Output leakage current. The current into the output of an amplifier with the output in the high level (off) state.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3).

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and

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may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	LM 139
02	LM 193

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC
Air Force - 19

Civil Agencies Coordinating Activities:
NASA - NA

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0817)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/112A	Microcircuits, Linear, Voltage Comparators, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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...these waveforms with the three-state output